ARJUNA JEE 2027

Lecture -06 | Mathematics

TRIGONOMETRIC FUNCTIONS

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Topics *to be covered*

A **Compound Angle Formula**

B **Question Practice**

C

D



RAPID FIRE ROUND

1. $\sin(131\pi - \theta) =$ _____.
2. $\cos\left(57\frac{\pi}{2} - \theta\right) =$ _____.
3. $\cos(82\pi - \theta) =$ _____.
4. $\cos\left(33\frac{\pi}{2} + \theta\right) =$ _____.
5. $\tan(183 \times 90^\circ + \theta) =$ $-\cot\theta$ _____.
6. $\tan(181 \times 90^\circ - \theta) =$ _____.
7. $\cot(57 \times 90^\circ + \theta) =$ _____.
8. $\sec\left(323\frac{\pi}{2} - \theta\right) =$ _____.
9. $\operatorname{cosec}(62\pi + \theta) =$ _____.
10. $\operatorname{cosec}\left(\theta - 71\frac{\pi}{2}\right) =$ _____.

(10)

$\operatorname{cosec}\left(\theta - 71\frac{\pi}{2}\right)$

$\downarrow$

$\operatorname{cosec}\left(-\left(71\frac{\pi}{2} - \theta\right)\right)$

(III)

$-\operatorname{cosec}\left(71\frac{\pi}{2} - \theta\right)$

$\downarrow$

$-\left(-\sec(\theta)\right) = \sec\theta$

Question Practice

Acidic Question

$$\tan\left(\underset{\text{II}}{\pi-\theta}\right) = -\tan\theta.$$

Evaluate :

(i) $$\frac{\sin\frac{\pi}{5}\cos\frac{7\pi}{9}\tan\frac{6\pi}{11}}{\cos\frac{2\pi}{9}\sin\frac{4\pi}{5}\tan\frac{5\pi}{11}}$$

$\cos\frac{7\pi}{9} = -\cos\frac{2\pi}{9}$, $\tan\frac{6\pi}{11} = -\tan\frac{5\pi}{11}$, $\sin\frac{4\pi}{5} = \sin\frac{\pi}{5}$

$= -1 \times -1 = +1$

(ii) $$\frac{\sin\frac{\pi}{7}\cos\frac{5\pi}{11}\tan\frac{3\pi}{7}}{\cos\frac{5\pi}{14}\sin\frac{\pi}{22}\tan\frac{4\pi}{7}} = -1$$

$\cos\left(\frac{\pi}{2}-\frac{\pi}{7}\right) = \sin\frac{\pi}{7}$

$\tan\frac{4\pi}{7} = -\tan\frac{3\pi}{7}$

$\cos\frac{5\pi}{14} = \cos\left(\frac{\pi}{2}-\frac{\pi}{22}\right) = +\sin\frac{\pi}{22}$

$\cos\frac{5\pi}{11} = \cos\left(\frac{\pi}{2}-\frac{\pi}{22}\right) = +\sin\frac{\pi}{22}$

$$\# \quad \frac{\pi}{2}-\frac{5\pi}{11} = \frac{11\pi-10\pi}{22} = \frac{\pi}{22}$$

Sachin Sir Special Summary (S^4):

$$\sin(A+B) = \sin A\cos B + \cos A\sin B$$

$$\sin(A-B) = \sin A\cos B - \cos A\sin B$$

$$\cos(A+B) = \cos A\cos B - \sin A\sin B$$

$$\cos(A-B) = \cos A\cos B + \sin A\sin B$$

(1) $\tan(A+B) = \left(\frac{\tan(A)+\tan(B)}{1-\tan A\tan B}\right)$

(2) $\tan(A-B) = \left(\frac{\tan A-\tan B}{1+\tan A\tan B}\right)$

(3) $\cot(A+B) = \left(\frac{\cot A\cot B-1}{\cot A+\cot B}\right)$

(4) $\cot(A-B) = \left(\frac{\cot A\cot B+1}{\cot B-\cot A}\right)$

$$\tan\left(\frac{\pi}{4}+\theta\right) = \left(\frac{1+\tan\theta}{1-\tan\theta}\right)$$

$$\tan\left(\frac{\pi}{4}-\theta\right) = \left(\frac{1-\tan\theta}{1+\tan\theta}\right)$$

Question

Prove that: $\frac{\sin(A-B)}{\cos A\cos B}+\frac{\sin(B-C)}{\cos B\cos C}+\frac{\sin(C-A)}{\cos C\cos A}=0$

$\frac{\sin A\cos B}{\cos A\cos B}-\frac{\cos A\sin B}{\cos A\cos B}$

$\tan A-\tan B+\tan B-\tan C+\tan C-\tan A=0$ ✓

Question **# Chota sa use:**

Finding value at (15°, 75°, 105°)

90° + 15°

$$\# \sin(15^\circ) = \sin(\overset{A}{45^\circ} - \overset{B}{30^\circ}) = \sin45^\circ\cos30^\circ - \cos45^\circ\sin30^\circ$$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2} = \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}$$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}}$$

$$\cos15^\circ = \cos(45^\circ - 30^\circ) = \cos45^\circ\cos30^\circ + \sin45^\circ\sin30^\circ$$

$$= \frac{\sqrt{3}+1}{2\sqrt{2}}$$

$\tan(75^\circ) = \tan(45^\circ + 30^\circ)$

(45° = A, 30° = B)

$$= \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ}$$

$$= \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} = \frac{\sqrt{3}+1}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{(\sqrt{3}+1)^2}{(\sqrt{3})^2 - 1^2}$$

$$= \frac{3+1+2\sqrt{3}}{3-1} = \frac{4+2\sqrt{3}}{2}$$

$$= 2+\sqrt{3}.$$

$\sin(105^\circ) = \sin(90^\circ + 15^\circ)$

$= +\cos 15^\circ$

REMEMBER AS RESULT (RAR)

#RAR

- $\sin(15°) = \sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}} = \cos(75°) = \cos\frac{5\pi}{12}$
- $\sin(75°) = \sin\frac{5\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}} = \cos(15°) = \cos\frac{\pi}{12}$
- $\tan(15°) = \tan\frac{\pi}{12} = 2-\sqrt{3} = \cot(75°) = \cot\frac{5\pi}{12}$
- $\tan(75°) = \tan\frac{5\pi}{12} = 2+\sqrt{3} = \cot(15°) = \cot\frac{\pi}{12}$

$\cos(75°) = \cos(90°-15°)$

$= +\sin 15°$

Ques:-

Quad. with rational coeff. whose one root is $\tan\frac{\pi}{12}$.

$x^2 - (\ 4\ )x + 1 = 0$

$\alpha = 2 - \sqrt{3}$.

$\hookrightarrow 2 + \sqrt{3}$

Question [JEE Repeating]

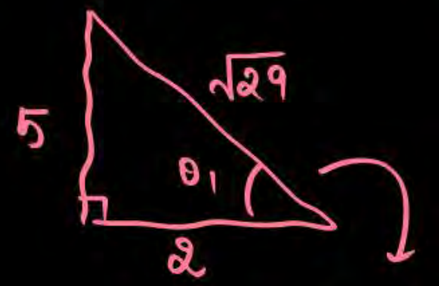
If $\sin(\overbrace{A-B}^{\theta_2}) = \frac{1}{\sqrt{10}}$, $\cos(\underbrace{A+B}_{\theta_1}) = \frac{2}{\sqrt{29}}$, find the value of $\tan 2A$ where A and B lie between 0 and $\frac{\pi}{4}$.

$$\tan\left(\overbrace{(A+B)+(A-B)}^{2A}\right) = \frac{\tan(A+B)+\tan(A-B)}{1-\tan(A+B)\tan(A-B)}$$

($\theta_1 = A+B$, $\theta_2 = A-B$)

$$= \frac{5/2 + 1/3}{1 - 5/2 \cdot 1/3} = \frac{17}{6 \times \frac{1}{6}} = 17$$

(I) $\cos(A+B) = \frac{2}{\sqrt{29}} = \frac{B}{H}$



$\tan\theta_1 = +\frac{5}{2}$

$\tan(A+B) = \frac{5}{2}$

$0 < A, B < \frac{\pi}{4}$

$$0 < A < \frac{\pi}{4}$$

$$0 < B < \frac{\pi}{4}$$

add

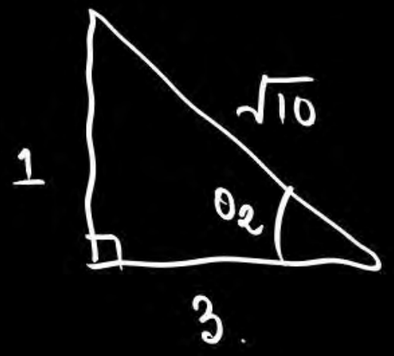
$$0 < (A+B) < \frac{\pi}{2}$$

θ_1

$\times(-1)$

$$0 > -B > -\frac{\pi}{4}$$

$$\sin(\overset{\theta_2}{A-B}) = \frac{1}{\sqrt{10}} = \frac{P}{H}$$



$$\tan(\theta_2) = \tan(A-B) = +\frac{1}{3}$$



$$0 < A < \frac{\pi}{4}$$

$$-\frac{\pi}{4} < -B < 0$$

add

$$-\pi/4 < A-B < \frac{\pi}{4}$$

$0 < A, B < \frac{\pi}{4}$

$$0 < A < \frac{\pi}{4}$$

$$0 < B < \frac{\pi}{4}$$

add

$$0 < (A+B) < \frac{\pi}{2}$$

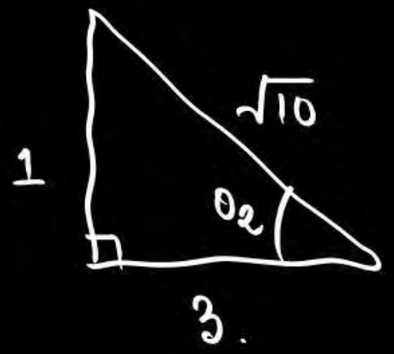
θ_1

$\times(-1)$

$$0 > -B > -\frac{\pi}{4}$$

θ_2

$$\sin(A-B) = \frac{+1}{\sqrt{10}} = \frac{P}{H}$$



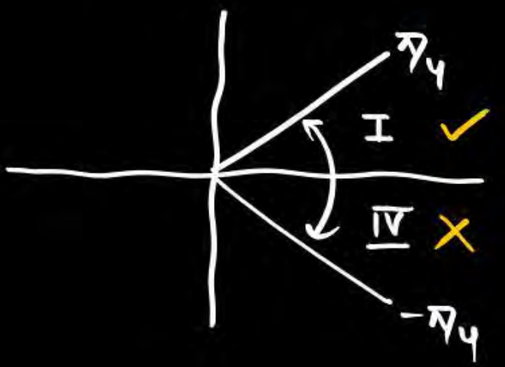
$$\tan(\theta_2) = \tan(\underbrace{A-B}_{I}) = +\frac{1}{3}$$

$$0 < A < \frac{\pi}{4}$$

$$-\frac{\pi}{4} < -B < 0$$

add

$$-\pi/4 < A-B < \frac{\pi}{4}$$



Question [JEE Repeating]

If $\sin(\overbrace{A-B}^{\theta_2}) = \frac{1}{\sqrt{10}}$, $\cos(\underbrace{A+B}_{\theta_1}) = \frac{2}{\sqrt{29}}$, find the value of $\tan 2A$ where A and B lie between 0 and $\frac{\pi}{4}$.

$$\# \quad \tan(2B) = \tan\left((\overset{\theta_1}{A+B}) - (\overset{\theta_2}{A-B})\right)$$

$$= \tan(\theta_1 - \theta_2)$$

$$= \frac{5/2 - 1/3}{1 + \frac{5}{2} \times \frac{1}{3}} = \frac{13}{11} \checkmark$$

Question [JEE Mains-2019]

If $\cos(\alpha+\beta)=\frac{3}{5}, \sin(\alpha-\beta)=\frac{5}{13}$ and $0<\alpha,\beta<\frac{\pi}{4}$ then $\tan(2\alpha)$ is equal to

A $\frac{21}{16}$

B $\frac{33}{52}$

C $\frac{63}{16}$

D $\frac{63}{52}$

DIBY - (X)

Question

Value of : $(\tan 100^\circ + \tan 125^\circ + \tan 100^\circ \tan 125^\circ)$ is

A 0

B −1

C 2

D 1

Sin & Cos ✓

$$\frac{\sin 100^\circ}{\cos 100^\circ} + \frac{\sin 125^\circ}{\cos 125^\circ} + \frac{\sin 100^\circ \sin 125^\circ}{\cos 100^\circ \cdot \cos 125^\circ}$$

$225^\circ = 45^\circ \times 5$

$= 180^\circ + 45^\circ$

$$\frac{\overset{A}{\sin 100^\circ}\overset{B}{\cos 125^\circ} + \overset{B}{\sin 125^\circ}\overset{A}{\cos 100^\circ} + \sin 100^\circ \sin 125^\circ}{\cos 100^\circ \cos 125^\circ}$$

$$\frac{\sin(100^\circ + 125^\circ) + \sin 100^\circ \sin 125^\circ}{\cos 100^\circ \cos 125^\circ}$$

Question

Value of : $(\tan 100^\circ + \tan 125^\circ + \tan 100^\circ \tan 125^\circ)$ is

- A 0
- B −1
- C 2
- D 1

$100^\circ + 125^\circ = 225^\circ$.

Applying 'tan' both side

$\tan(\underset{A}{100^\circ} + \underset{B}{125^\circ}) = \tan 225^\circ$

$$\frac{\tan 100^\circ + \tan 125^\circ}{1 - \tan 100^\circ \tan 125^\circ} = 1.$$

$\tan 100^\circ + \tan 125^\circ = 1 - \tan 100^\circ \tan 125^\circ$.

$\tan(180^\circ + 45^\circ) = +\tan 45^\circ$

$= 1$

SAS

For tan & cot wale Ques^n.

⇓

Start with angles

Question

Prove that : $\tan 80^\circ = \tan 10^\circ + 2\tan 70^\circ$

$\tan(90^\circ - 10^\circ) = \cot 10^\circ$

$= \frac{1}{\tan 10^\circ}$

$70^\circ + 10^\circ = 80^\circ$

Apply tan

$\tan 70^\circ + \tan 10^\circ = \tan 80^\circ$

$70^\circ = 80^\circ - 10^\circ$

$\tan 70^\circ = \tan(80^\circ - 10^\circ)$

$= \frac{\tan 80^\circ - \tan 10^\circ}{1 + \tan 80^\circ \tan 10^\circ}$

$\frac{1}{1} \rightarrow 2$

$2\tan 70^\circ = \tan 80^\circ - \tan 10^\circ$

$\tan(80^\circ) = \tan(90^\circ - 10^\circ)$

$\tan(90^\circ - \theta) = \cot\theta$.

$= \cot 10^\circ$

$\tan(80^\circ) = \frac{1}{\tan 10^\circ}$

$\tan 10^\circ \tan(80^\circ) = 1$.

if $A + B = 90^\circ$

$\tan A \cdot \tan B = 1$ ✓

Question

Show that : $\tan 3A - \tan 2A - \tan A = \tan A \tan 2A \tan 3A$



$A + B = 45^\circ$

$\tan(A+B) = \tan 45^\circ$

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = 1$$

$$\tan A + \tan B = 1 - \tan A \tan B$$

$$1 + \tan A + \tan B + \tan A \tan B = 1 + 1$$

$$(1 + \tan A)(1 + \tan B) = 2$$

\# RAR

↳ factorisation

$$1 + x + y + xy = (1+x)(1+y)$$

$(1+x) + y(1+x)$

$A + B = 45^\circ$

$(1 + \tan A)(1 + \tan B) = 2$.

$\tan 7^\circ = \tan(90^\circ - 83^\circ)$

$= +\cot 83^\circ$

$(1 + \tan 7^\circ)(1 + \tan 38^\circ) = 2$ ✓

$(1 + \cot 83^\circ)(1 + \tan 38^\circ) = 2$

$(1 + \cot 83^\circ)(1 + \cot 52^\circ) = 2$

$(1 + \tan 221^\circ)(1 + \tan 4^\circ) = 2$, $A + B = 225^\circ$.

$A + B = 225^\circ$

$180^\circ + 45^\circ$

$\tan(A+B) = \tan 225^\circ$

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = 1$$

$$\tan A + \tan B = 1 - \tan A \tan B$$

$$1 + \tan A + \tan B + \tan A \tan B = 1 + 1$$

$$(1 + \tan A)(1 + \tan B) = 2.$$

* RAR

factorisation

$$1 + x + y + xy = (1+x)(1+y)$$

$(1+x) + y(1+x)$

Question

If $A + B = 45°$, then prove that : $(\cot A - 1)(\cot B - 1) = 2$



Question [JEE Mains-2022]

If $\cot\alpha = 1$ and $\sec\beta = -\frac{5}{3}$, where $\pi < \alpha < \frac{3\pi}{2}$ and $\frac{\pi}{2} < \beta < \pi$, then the value of $\tan(\alpha + \beta)$ and the quadrant in which $\alpha + \beta$ lies, respectively are:

A $-\frac{1}{7}$ and IVth quadrant



B 7 and Ist quadrant

C -7 and IVth quadrant

D $\frac{1}{7}$ and Ist quadrant

DO IT BY YOURSELF (DIBY)

DIBY-08

Prove that : $\frac{\sin A}{1+\cos A}+\frac{1+\cos A}{\sin A}=2\operatorname{cosec} A$

DIBY-09

Prove that : $\sqrt{\frac{1-\sin A}{1+\sin A}}=\sec A-\tan A$

DIBY-10

Prove that : $\frac{\operatorname{cosec} A}{\operatorname{cosec} A-1}+\frac{\operatorname{cosec} A}{\operatorname{cosec} A+1}=2\sec^2 A$

DIBY-11

Prove that : $\frac{\operatorname{cosec} A}{\cot A+\tan A}=\cos A$

DIBY-12

Find value of following

(i) $\sin^2\frac{\pi}{6}+\cos^2\frac{\pi}{3}-\tan^2\frac{\pi}{4}=$

(ii) $2\sin^2\frac{\pi}{6}+\operatorname{cosec}^2\frac{7\pi}{6}\cos^2\frac{\pi}{3}=$

(iii) $\cot^2\frac{\pi}{6}+\operatorname{cosec}\frac{5\pi}{6}+3\tan^2\frac{\pi}{6}=$

(iv) $2\sin^2\frac{3\pi}{4}+2\cos^2\frac{\pi}{4}+2\sec^2\frac{\pi}{3}=$

[Ans. : (i) $\frac{-1}{2}$, (ii) $\frac{3}{2}$, (iii) 6, (iv) 10]

DIBY-13

$\frac{\tan(90°-\theta)\sec(180°-\theta)\sin(-\theta)}{\sin(180°+\theta)\cos(360°+\theta)\operatorname{cosec}(90°-\theta)}$ is equals

(A) $\operatorname{cosec}\theta$ (B) 1 (C) –1 (D) $-\operatorname{cosec}\theta$

[Ans. : (D)]

DIBY-14

$\sec(270^\circ - A)\sec(90^\circ - A) - \tan(270^\circ - A)\tan(90^\circ + A)$

[Ans. : –1]

DIBY-15

$\cot A + \tan(180^\circ + A) + \tan(90^\circ + A) + \tan(360^\circ - A)$

[Ans. : 0]

DIBY-16

$$\frac{\tan(180^\circ-\theta)\cos(180^\circ-\theta)\tan(90^\circ-\theta)}{\sin(90^\circ+\theta)\cot(90^\circ-\theta)\tan(90^\circ+\theta)}$$

[Ans. : –1]

DIBY-17

If $\tan A = 2$ and $A \in \left(\pi, \frac{3\pi}{2}\right)$ then the expression $\frac{\cos A}{\sin^3 A + \cos^3 A}$ is equal to

[Ans. : 5/9]

DIBY-18

$\sin^2\frac{\pi}{18}+\sin^2\frac{\pi}{9}+\sin^2\frac{7\pi}{18}+\sin^2\frac{4\pi}{9}$ equal

(A) 0 (B) 1 (C) –1 (D) 2

[Ans. : (D)]

DIBY-19

Prove that

1. $\tan 720^\circ-\cos 270^\circ-\sin 150^\circ\cos 120^\circ=\frac{1}{4}$

2. $\sin 780^\circ\sin 480^\circ+\cos 120^\circ\sin 150^\circ=\frac{1}{2}$

DIBY-20

Prove that $\sec\left(\frac{3\pi}{2}-\theta\right)\sec\left(\theta-\frac{5\pi}{2}\right)+\tan\left(\frac{5\pi}{2}+\theta\right)\tan\left(\theta-\frac{3\pi}{2}\right)=-1.$

DIBY-21

If $\sin x = -\frac{3}{5}$ where $\pi < x < \frac{3\pi}{2}$, then $80(\tan^2 x - \cos x)$ is equal to

(A) 109 (B) 108 (C) 19 (D) 18

[Ans. : (A)]

DIBY-22

1. $\sin(405^\circ)$ [Ans. : $\frac{1}{\sqrt{2}}$]
2. $\sec(-1470^\circ)$ [Ans. : $\frac{2}{\sqrt{3}}$]
3. $\tan(-300^\circ)$ [Ans. : $\sqrt{3}$]
4. $\cot(585^\circ)$ [Ans. : 1]
5. $\operatorname{cosec}(-750^\circ)$ [Ans. : -2]
6. $\cos(-2220^\circ)$ [Ans. : 1/2]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians